

Q. No.	Questions
UNIT 1	SECTION 1
1	A software company tests the following algorithm before integrating it into a banking application. During code review, the developer suspects that the algorithm contains logical errors that may produce incorrect results while identifying the customer with the highest transaction amount. Analyze the given algorithm and provide the necessary corrections, the output for A = 18, B = 27, C = 15, and the reason the corrected algorithm produces the expected result. Start Read A, B, C if A>B and A>C Largest = A else if B<A and B>C Largest = B else Largest = A Print Largest Stop (5 Marks)
2	A customer-service kiosk implements a waiting line using collections.deque, following the FIFO (First-In, First-Out) principle. Customers report that the wrong person is being served first. Interpret the code against the FIFO principle, identify the faulty line, explain why it produces incorrect output, and rewrite the corrected code with its final traced output. <pre> from collections import deque queue = deque() queue.append(1) queue.append(11) queue.append(15) front = queue.pop() # serves the next customer print("Served:", front) print("Remaining queue:", list(queue)) </pre> (5 Marks)
3	A web browser uses a Stack to store the pages visited by a user. The user visits three pages Google, YouTube, and Wikipedia in that order. The browser then performs one POP operation followed by a PEEK operation. Analyze the Stack operations and show the Stack contents after each step. (5 Marks)
4	(i) A navigation software company is evaluating multiple route-planning algorithms for a map application. Since the application will be deployed on different devices, the developers require a hardware-independent method to compare algorithm performance before implementation. During testing, one route-planning module searches for a destination by checking each route sequentially until the required destination is found at the end of the list. Analyze the significance of asymptotic notations in algorithm analysis for comparing the route-planning

	algorithms. (3 Marks)
	(ii) Identify the appropriate asymptotic notation for the given sequential search operation and justify the selection. (2 Marks)
UNIT 1	SECTION 2
5	An online banking system maintains a set of active customer usernames. When users attempt to log in, the security system must verify whether their username exists in the database. Only registered active customers are granted access. GIVEN DATA: ActiveUsers = {"Asha", "Kiran", "Mohan", "Priya", "Ravi"} Login Attempts: "Priya" and "Suresh" Using set membership operators, determine whether each login request should be granted or denied. For each user: (a) Check whether "Priya" exists in the ActiveUsers set (b) Determine login status for Priya with justification (c) Check whether "Suresh" exists in the ActiveUsers set (d) Determine login status for Suresh with justification (e) Explain why membership operators are suitable for this security application" (5 Marks)
6	A cloud storage company maintains 8 million customer records in sorted order. The current system uses Linear Search, but the software architect proposes replacing it with Binary Search because the database is expected to grow rapidly over the next few years. Determine both searching techniques and recommend the most suitable algorithm for the company. Justify your recommendation using their asymptotic time complexities (5 Marks)
7	An online shopping website receives nearly one million product IDs every day from multiple vendors. Many vendors upload the same product repeatedly. One software engineer suggests checking every new product against all existing records manually, while another recommends storing the IDs in a Python set. Compare the two approaches and determine which method should be adopted for handling duplicate products efficiently. Justify your answer. (5 Marks)
8	(i) An online shopping application stores active coupon codes in a Python set. A customer tries to remove the coupon code 'EXPIRED99', but it is already been removed earlier. The application should continue running without generating an error when the coupon code is not found. coupons = {'SAVE10', 'WELCOME5', 'FESTIVE20'} Implement the appropriate Python set method to remove the coupon code 'EXPIRED99' and determine the output. (3 Marks) (ii) Compare the discard() and remove() methods, and identify why discard() is more suitable for this situation. (2 Marks)
UNIT 2	SECTION 1
9	A university examination cell stores 2,00,000 student records in a sorted database. One software team proposes checking records one by one, while another team suggests repeatedly dividing the search space

	into halves. The examination controller wants a solution that remains efficient even when the database size doubles every year. Examine the two approaches and determine which algorithm should be selected. Justify your answer using asymptotic growth rates and scalability considerations. (5 Marks)
10	An online shopping company guarantees that its recommendation engine will never take more than a certain amount of time to generate suggestions, regardless of customer behavior. During a technical review, engineers debate whether best-case, average-case, or worst-case analysis should be reported to management. Classify the three types of complexity analysis and recommend the most appropriate measure for providing such a guarantee. (5 Marks)
11	A cloud service provider guarantees customers that its response time will never exceed a specified limit regardless of workload. During a review meeting, engineers debate whether best-case, average-case, or worst-case analysis should be reported. Evaluate the three forms of complexity analysis and recommend the most suitable one for providing such a guarantee. (5 Marks)
12	(i) A software company is developing a financial analytics application that processes daily profit and loss values for thousands of transactions. Two software engineers are discussing the implementation of the Maximum Subsequence Sum (MSS) problem for analyzing the maximum cumulative profit. Engineer A recommends Algorithm 1 ($O(N^2)$), stating that it is simple and sufficiently efficient. Engineer B argues that Algorithm 2 ($O(N)$) should be used because the application must process approximately 50,000 transactions every day with minimal execution time. The project manager asks both engineers to justify their recommendations before selecting the final algorithm Algorithm 2: Maximum Subsequence Sum (Linear-Time Algorithm) 1. maxSum $\leftarrow$ 0 2. thisSum $\leftarrow$ 0 3. for i $\leftarrow$ 0 to N-1 do 4. thisSum $\leftarrow$ thisSum + A[i] 5. if thisSum > maxSum then 6. maxSum $\leftarrow$ thisSum 7. else if thisSum < 0 then 8. thisSum $\leftarrow$ 0 9. return maxSum . Apply Algorithm 2 (Linear-Time MSS Algorithm) to the input sequence Trace the values of the running sum and maximum sum after processing each element, and determine the final Maximum Subsequence Sum. (3 Marks) (ii) Compare the execution strategies of Algorithm 1 ($O(N^2)$) and Algorithm 2 ($O(N)$) for solving the Maximum Subsequence Sum (MSS) problem. Based on their time complexities, recommend the most suitable algorithm for processing N = 50,000 integers with justification. (2 Marks)
UNIT 2	SECTION 2

13	An if/else block in the record-validation code checks a condition and executes a fast $O(1)$ branch if true, or a slow $O(N)$ branch (looping through all records) if false. Derive the best case and worst case and explain under what condition the Average Case would need probability theory instead of a simple formula (5 Marks)
14	For instant lookup of student records by roll number, the team implements a hash table using separate chaining (open hashing) with hash function $h(K) = K \bmod m$. Determine the phenomenon that occurs when two roll numbers are mapped to the same hash table cell, and explain how increasing the load factor () affects the efficiency of search operations. (5 Marks)
15	The team builds a keyword search tool to find a specific word (pattern) inside long alumni transcripts (text). They are deciding between Brute-Force matching and Horspool's Algorithm. Explain the key difference in their approach and state why Horspool's is generally faster in practice despite having the same worst-case complexity (5 Marks)
16	(i) A software developer writes two nested loops to process an image of size $N \times N$ pixels. Analyze the time complexity of the nested-loop program. (3 Marks)
	(ii) Evaluate why nested loops require multiplication of loop sizes during complexity analysis. (2 Marks)
UNIT 3	SECTION 1
17	A software company is developing an Undo feature for its text editor. During testing, a newly hired engineer suggests implementing the feature using a Queue instead of a Stack, arguing that both data structures store data sequentially. Before finalizing the design, the development team requests a technical justification for selecting the appropriate Abstract Data Type (ADT). As a software engineer, recommend the most appropriate Abstract Data Type for implementing the Undo feature of the text editor. Support your recommendation by describing the characteristics of the selected ADT and explaining how its operations satisfy the application's functional requirements. (5 Marks)
18	An event-management application must satisfy two competing requirements. First, volunteers must be added to or removed from either the beginning or the end of a duty list as last-minute changes occur. Second, urgent tasks such as a reported stage malfunction must always be handled before routine tasks, such as refilling water stations, regardless of which was logged first. Recommend which queue variant should be used for each requirement and justify your recommendation for each based on their operational rules. (5 Marks)
19	A software team at a warehouse management company is redesigning its inventory audit module. The lead architect must decide whether the shelf-by-shelf stock counting routine should use a looping mechanism or a self-calling function, given that shelves are nested within storage bays that are themselves nested within other bays to an unknown depth. Justify, why a self-calling function is better suited than a loop when the nesting depth of a data structure is unknown in advance. (5 Marks)
20	(i) A circular queue of size 5 (indices 0 to 4) is used to manage boarding passes at an airport gate counter, with $\text{front} = \text{rear} = -1$ initially. Passengers are enqueued in the order P1, P2, P3, P4, P5, completely filling the queue, after which two passengers, P1 and P2, are dequeued in sequence. Compute the resulting front and rear index values after the two dequeue operations described above.

	(3 Marks)
	(ii) Predict what happens if the counter now attempts to enqueue two new passengers, P6 and P7, one after another, and state which of the two insertions (if any) succeeds. (2 Marks)